Joseph Poon

Bio

Technical Lead for brwne & Software Engineering student at SIT with a passion for high-performance systems. CLI user, software architect & open-source developer

Contact

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Education

BEng (Hons) in ICT (Software Engineering)
SIT (Singapore Institute of Technology) Aug 2025 – Aug 2029

Diploma in Cybersecurity & Digital Forensics
Nanyang Polytechnic *Mar 2020 – Mar 2023*

- GPA: 3.33
- **Director's List (AY 2022/2023):** Awarded to top 15% of cohort for academic excellence in a semester
- **Good Progress Award (2022):** Awarded to top 10% of students in terms of improvement of academic performance & good conduct
- **Diploma-Plus Modules:** Cloud Computing Foundation, CEH (Certified Ethical Hacker)

Technical Skills

- Languages & Systems Programming
  - Go, Python, JavaScript, TypeScript, Rust
- OS & Scripting: Linux (Debian), Bash
- DevOps Methodologies: Agile, Scrum
- Container Orchestration
  - Docker, Kubernetes (Helm, Istio, Kyverno, Knative)
- CI/CD: Jenkins, ArgoCD

Working Experience

Acting CTO
PirateShip *Dec 2025 – Present*
Technical Lead *5 mths | Aug 2025 – Dec 2025*
Intern *3 mths | May 2025 – Jul 2025*

- **brwne:** Adversarial, multi-agentic QA platform to generate, maintain, & validate E2E test suites
  - Engineered an iterative refinement method for generating Playwright tests
  - Prototyped a self-healing mechanism for differentiating intentional code changes & regressions
- **brwne legacy:** A real-time collaborative tool to prevent merge conflicts for dev teams
  - Developed a Rust desktop client via Tauri
  - Engineered a high-performance Go local daemon sidecar, monitoring Git & file events in real time
  - Deployed CI/CD pipeline for backend server onto Linode & Linux release
  - Achievements:
    - NUS VIP (Venture Initiation Programme) Awardee: $10,000 *May 2025*
    - Best Project Award for NUS STePS: $2,000 *Apr 2025*

Tax Officer
IRAS *5 mths | Feb 2023 - Jun 2023*

- Aided in filing of several tax forms, including Individual Income Tax (Form B, B1, M), Partnerships (Form P) & Stamp Duty (Lease & Tenancy Agreement) for taxpayers
- Setup of Linux systems on IoT devices & printers for queue display, developed a website for printing queue numbers

Software Engineering Intern
SAP *6 mths | Mar 2022 - Aug 2022*

- Developed an automated framework for gathering statistics on integration & unit tests
- Implemented HA (High Availability) for Kubernetes components
- Performed pipeline migration for outdated Jenkins pipelines
- Contributed to feature development, testing, deployments & releases in an Agile Scrum team

Competitions

HackRift 2025
SIT — *1st Place | Dec 2025*

- Developed `TriageX`: an autonomous voice-first medical triage kiosk powered by a multi-agent AI ensemble designed to slash polyclinic wait times
- Won 1st place with a $4,000 prize of 150+ participants

Cursor Hackathon Singapore
Cursor — *Oct 2025*

- Developed `Lumina`: a real-time AI agent that facilitates classroom discussions with voice interventions, keeping students engaged
- Competed against 1,000+ participants with a $100k+ prize pool

IEEE SOLI 2025 Hackathon
IEEE — *3rd Place | Sept 2025*

- Developed `err`: an AI supply chain anomaly detection web app
- Won 3rd place with a $900 prize pool

Code Overflow
NYP — *1st Place | Sept 2022*

- Sustainability-themed coding hackathon conducted by Nanyang Polytechnic School of Information Technology club
- Developed `EnergyClash`: a web app targeted at reducing household electricity consumption via gamification

Debug Minds Hackathon
NYP — *2nd Place | Sept 2021*

- Creative coding hackathon conducted by NYP School of Information Technology club

Certifications

Google IT Automation with Python
Google — *Mar 2023*

- Completion of 6 courses: Crash Course on Python, Using Python to Interact with the OS, Introduction to Git & Github, Troubleshooting & Debugging Techniques, Configuration Management & the Cloud, Automating Real-World Tasks with Python

Alibaba Cloud Associate (ACA)
Alibaba Group — *Oct 2022*

- Certifies the proficient use of Alibaba Cloud Computing Products.

Certified Ethical Hacker (CEH v11)
EC-Council — *April 2021*

- Demonstrates ability to identify & mitigate security risks.

Projects / Open Source

Life OS
Jul 2025 – Present

- Life management system to organize & manage your life as an Obsidian vault
- Developed dynamic dashboards to manage tasks, projects, & journals using Dataview & CustomJS plugins to create a centralized personal workspace

cozydot
Jul 2024 – Present

- cozydot is an automated post-install, update, & config (dotfile) manager for Linux
- Uses declarative YAML configs to manage system packages (apt, Flatpak, Cargo), dotfiles (via GNU Stow), & application settings for a customizable & idempotent setup

Sustainacity
Dec 2022

- AR (Augmented Reality) enhanced web app to showcase DBS' sustainability initiative. Created for NYP x DBS AR Sustainability Storytelling Competition 2022

adore_melxdy
Sept 2021

- Cloud-hosted Youtube music bot for Discord

–vouch
Aug 2021

- Web app aimed at automation via data extraction & data matching for vouching in the audit industry. Created for AIC21

Soft Skills

- Technical Leadership
- Performance Under Pressure
- Systems Thinking
- Rapid Prototyping & Execution
- Adaptability & Fast Learning